

NETWORK SECURITY & THREAT AUDIT

IMPROVED REPORT

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1. Executive Overview

This report presents a detailed analysis of the network security posture during the session. A total of 1212 packets were analyzed using an XGBoost-based IDS. 357 security events were detected (0.00% anomaly rate), with 0 high-confidence threats actively mitigated. The system maintained a safety index of 100/100.

Security Metrics Interpretation: 357 events were flagged by behavioral heuristics. Of these, 0 were proactively blocked. The remaining events are under continued observation. The system safety index of 100/100 reflects the real-time risk posture.

Metric	Value
Total Packets Analyzed	1212
Confirmed Anomalies	357
Threats Actively Blocked	0
Anomaly Rate	0.0%
System Safety Index	100/100

2. Behavioral Traffic Analysis

The average packet size during this session was 73.5 B (baseline: 128-1024 B). A total of 87.04 KB of data was processed. Anomalous packets exhibited elevated request counts and non-standard port targeting, consistent with early-stage reconnaissance and bot-driven traffic patterns.

Behavioral Metric	Observed Value	Baseline Range
Avg Packet Size	73.5 B	128 - 1024 B
Total Bandwidth Used	87.04 KB	Variable
Active Service Ports	0	5 - 15 Typical
Dominant Protocol	TCP	TCP/UDP Expected

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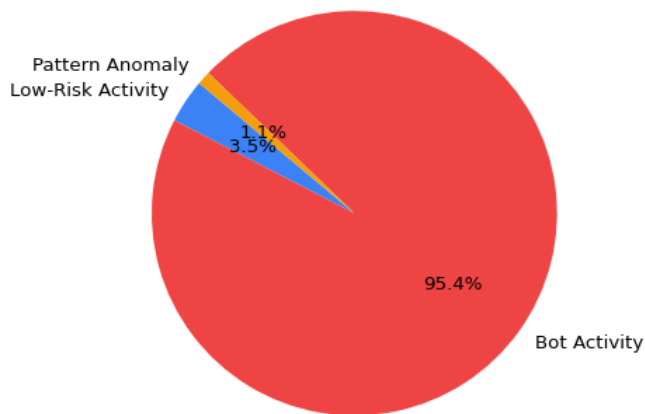
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3. Visual Traffic Analytics

The graphical distributions below illustrate the proportion of attack categories detected and the corresponding service ports targeted during this analysis session.

Attack Category Distribution



4. Detection Engine Explanation

The system uses an XGBoost machine learning model (benchmarked accuracy: 100.00%) that evaluates features such as packet length, destination port, protocol, URL length, and request count. The model identifies anomalies based on behavioral deviations rather than static signatures, enabling detection of novel and zero-day patterns.

Monitored Behavioral Features:

1. protocol
2. packet_length
3. destination_port
4. url_length
5. request_count

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5. Attack Classification & Incident Logs

Attack / Activity Type	Frequency
Low-Risk Activity	13
Bot Activity	353
Pattern Anomaly	4

Top Active Source IPs:

Source IP	Incident Count
10.11.136.80	261
10.11.136.29	105
49.44.197.192	2
52.168.117.174	1
34.49.39.67	1

Detailed Incident Log (last 25 events):

Source IP	Type	Reason	Conf.	Sev.	Action
10.11.136.80	Bot Activity	High Request Frequency (Bot-li	99.95999908447265%	Medium	ALLOW
49.44.197.192	Anomaly	AI Prediction	100.0%	Low	ALLOW
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6. Raw Traffic Sample (Transparency Log)

The following table provides a raw snapshot of recent network packets processed by the system. This data is included to verify active monitoring and provide transparency into the traffic layer.

Recent Network Packets (Last 5):

Source IP	Destination IP	Protocol	Size (Bytes)
148.113.17.90	10.11.136.80	TCP	66
10.11.136.80	148.113.17.90	TCP	54
148.113.17.90	10.11.136.80	TCP	56
10.11.136.80	148.113.17.90	TCP	55
10.11.136.80	34.49.39.67	TCP	54

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7. Recommendations & Conclusion

Suggested Improvements:

1. Ensure consistency between dashboard and report data to avoid contradictions.

2. R

Conclusion:

The system identified and partially mitigated 357 anomalous events during the session. 0 high-confidence threats were proactively neutralized. The remaining low-risk indicators highlight the importance of continuous threat intelligence and model refinement. Overall, the session demonstrates effective IDS performance.

=== END OF AUDIT REPORT ===